David Z Osterman

(908) 821-6297 | Westfield, NJ | [Email] | [LinkedIn] | [Github] | [Website]

SUMMARY

PhD scientist with a proven track record of analytical prowess and data science expertise. My work experience demonstrates my strong aptitude for flexible thinking, communication of technical ideas to non-technical people, and a commitment to prioritizing team success. Seeking a position as a data scientist, machine learning engineer, or quantitative analyst.

SKILLS & CERTIFICATIONS

- Languages & Platforms: python, SQL, MySQL, Google Cloud, C, C++, git, bash, Linux
- Python Libraries: numpy, pandas, PyTorch, sklearn, tensorflow, NLTK, seaborn, matplotlib
- Machine Learning: ML Ops, NLP, deep learning, CNNs, Reinforcement Learning, Markov processes, Thompson sampling, UCB, data collection and cleaning, regression, classification, clustering, error analysis, API
- Quantitative: A/B testing, ETL, stochastic optimization, statistical analysis, data visualization, data pipelines
- Soft Skills: leadership, project management, communication skills, time management, flexibility, quick learner
- Certifications: The Erdős Institute Deep Learning Boot Camp (in-progress)

WORK EXPERIENCE

University of Massachusetts Amherst Physics Department: Amherst, MA **May 2021 - Feb 2024**

Graduate Research Assistant

- Led a team of three graduate students and one undergraduate student in an upgrade project resulting in a 200% increase in experiment performance.
- Collected, cleaned, and processed experiment data in python. Employed causal inference and predictive models, which uncovered a 150% to 350% increase in device performance.
- Performed time series and event-based analysis, which contributed to two papers, including a world-leading publication (currently in-preparation).

Argonne National Laboratory: Lemont, IL **June 2021 - Feb 2024**

SCGSR Fellow/CNM User

- Upgraded data acquisition system to be fully automated - which reduced required manpower to one person - and fully remote - which reduced required active time spent by 90%.
- Employed causal inference and predictive models for stochastic physical processes, and verified the effectiveness of a potential strategy for the collaboration.
- Designed, built, tested, and ran a novel experiment from scratch, resulting in a publication (currently in-preparation).

Brown University Physics Department: Providence, RI **Sep 2016 - Jan 2021**

Graduate Research Assistant

- Created the code basis (in C) of time series data digitization for the lab. Performed multivariate event ordering.
- Designed and ran tests verifying the effectiveness of critical sub-processes of the main data acquisition device in the lab.
- Studied and modeled the Brownian motion of stochastic particles.

SELECTED PROJECTS

Team Comets: Bodybuilder Judge [LINK] - *The Erdős Institute* **Fall 2024**

- My team employed two convolutional neural networks - one original and one ResNet50-based - that take images of two bodybuilders as inputs and judges which would beat the other in a competition. Our models were trained on over 11,000 image pairs from the results of bodybuilding competitions scraped from npcnewsonline.com.

Google Maps Re-Star [LINK] **Summer 2024**

- The goal of the project was to use Natural Language Processing to re-assign ratings out of five stars to restaurants based on the sentiment of the written reviews. I employed a pre-trained RoBERTa model to perform sentiment analysis on hundreds of reviews per restaurant.

EDUCATION

University of Massachusetts Amherst, *PhD* **2024**
Brown University, *Masters of Science* **2021**
Rutgers University, *Bachelor of Science in Physics and Mathematics* (GPA = 3.99) **2015**